Manjeet SINGH, 28

Ph.D: FOOD TECHNOLOGY (Final Year), FRANCE

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I am a final-year PhD researcher in Food Technology working on the valorisation of mangrove-derived resources and their impact on food quality. My research focuses on the extraction, physicochemical characterization, and functional evaluation of bioactive compounds, particularly polyphenols, to enhance oxidative stability and nutritional performance in food systems such as oils, seafood, and cereal-based products. Using advanced chromatographic and spectroscopic techniques, I investigate antioxidant mechanisms and food matrix interactions. With international experience across Europe, Africa, and India, I contribute to interdisciplinary research at UMRT BioEcoAgro. I am now seeking postdoctoral opportunities within my field, aiming to apply my expertise while continuing to learn and collaborate across disciplines.

Work Experience and International Mobilities

1. International Field Research – Mangrove Sampling (Supported by **IFSEA**)

University of Nairobi, Department of Biology, Kenya

Mombasa, Kwale, Diani, Shimoni | November 2025

- Conducted systematic field sampling of mangrove species (leaves and fruits) across coastal ecosystems for bioactive compound extraction.
- Performed species identification, sample preservation, and pre-processing for downstream analytical applications.
- Managed phytosanitary compliance, export documentation, and regulatory procedures for international sample transport.
- Coordinated field logistics and collaborated with local research teams.

2. International Field Research – Mangrove Sampling (Supported by **IFSEA**)

Federal University of Technology Akure (FUTA), Nigeria

Araromi Coastal Region | May 2025

- Executed field collection of mangrove plant materials for polyphenol extraction and food application studies.
- Ensured proper sampling protocols, preservation techniques, and traceability of collected materials.
- Completed phytosanitary procedures, documentation, and international sample shipment logistics.
- Collaborated with academic partners during field and pre-processing activities.

3. Research Intern – Food Biotechnology | (Masters Thesis)

University of Valencia, Spain | *Mar – Sep 2022*

- Developed fortified sourdough powder formulations for industrial evaluation and scale-up.
- Contributed to product development, quality assessment, and experimental protocol design.
- Assisted in laboratory operations under GMP/GLP standards.

4. Research Intern – Microbial Biotechnology

TERRA Research Center, Belgium | *Jul – Aug 2021*

- Investigated stress responses of *Saccharomyces cerevisiae* in controlled bioprocess conditions.
- Operated and monitored a 30 L laboratory-scale bioreactor.
- Supported experimental workflows including media preparation and culture handling.

5. Quality Control Intern – Beverage Production

Coca-Cola, India | *Jun – Jul 2019*

- Performed physicochemical quality analysis (Brix, pH, CO_2) and process monitoring.
- Gained practical exposure to HACCP, CIP systems, and industrial quality control protocols.

Education

1) Ph.D. in Food Technology | Université d'Artois, France | *2023–2026*

Topic: Valorisation of mangrove resources and their impact on food quality.

2. M.Sc. in Microbiology and Physicochemistry for Food and Wine Processes

University of Burgundy (AgroSup), France | *2020–2022*

- Erasmus+ Exchange: **University of Liège** (Agro-Biotech), Belgium | ***2021***
- Erasmus+ Thesis Internship: **University of Valencia** (AgroBiotech), Spain | ***2022***

3. B.Tech in Food Technology, India | ***2016–2020***

Shoolini University of Biotechnology and Management Sciences.

Projects

1. **Title – (Ph.D Thesis)** Contribution for the valorization of the resources from mangroves in food technology.

   Supervisor – Prof. Romdhane Karoui **Contact**: romdhane.karoui@univ-artois.fr
   Institution – Universite d'Artois, France (For reference)

2. **Title** – (**Master Thesis)** Use of dry sourdoughs inoculated with lactic acid bacteria, fermentation for bread biopreservation, Quality analysis and Regulations (2022).

   Supervisor – Dr. Giuseppe Meca, **Contact**: Giuseppe.Meca@uv.es
   Institution – University of Valencia, Spain (For reference)

3. **Title** – Food safety: Application of visible light and UV for the decontamination of food and surfaces against HuNoV and using MS2 Bacteriophage as its surrogate. (2021-2022).

   Supervisor – Prof Stephane Guyot **Contact:** stephane.guyot@agrosupdijon.fr
   Institution – PAM UMR lab, AgroSup Dijon, France (For reference)

4. **Title** - Bio-Processing, fermentation and verification of stress conditions on yeast (2021).

   Supervisor – Dr. Frank Delvigne
   Institution- University of liege, AGBT, Belgium.

5. **Title** – Alternate protein-To study the insect protein, product formulation and market evaluation. (2020-2021).

   Supervisor – Prof Camille Loupaic.
   Institution – University of Bourgogne, France.

6. **Title** – Microalgae, Spirulina and Algae "From Formulation to consumers"& Sensory Analysis.

   Supervisor – Prof Camille Loupaic.
   Institution – University of Bourgogne, France.

7. **Title** - Unconventional yeast strains having a positive impact on lowering the alcohol content of wines. **(2020) (Bachelors Thesis)**

   Supervisor – Prof. Somesh Sharma,
   Institution – Shoolini University of Biotechnology and management sciences. India

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Publications

Published

- Singh, M.; Botosoa, E.; Karoui, R. Monitoring of Antioxidant Efficacy of Mangrove-Derived Polyphenols in Linseed Oil by Physicochemical and Fluorescence Methods. **Antioxidants**, 2025, 14(2), 192.

  Link- https://doi.org/10.3390/antiox14020192

Under Review

- Innovative Use of Mangrove Leaf Extracts for Sea Bream (*Sparus aurata*) Fillet Preservation: Assessing Immersion and Alginate Encapsulation Approaches. Food Chemistry.
- Multimodal Spectroscopic and UHPLC Characterization of Mangrove Polyphenols and Their Role in Enhancing the Oxidative Stability of Linseed Oil. Food Chemistry.
- Integrated Targeted and Library-Assisted Profiling of Phenolic Compounds in Fractionated Sub-Saharan Mangrove Species Using GC–MS. Food Chemistry.
- Physicochemical Characterisation of Mangrove Leaf Powders and Rapid FT-MIR Monitoring of Linseed Oil Oxidative Stabilisation. Antioxidants.
- Potential Sources of Bioactive Compounds Extracted from Mangroves Towards Formulation of Healthy Food Products. Trends in Food science.
- Chromatographic Prowess: Unraveling Mangrove Bioactive Compounds Using HPLC and GC Analysis of Biomolecules. Journal of chromatography A.

Research and Technical skills

- **Analytical & Instrumentation Techniques**

  HPLC–MS (Shimadzu), GC–MS (Shimadzu), FTIR spectroscopy (Shimadzu), Fluorescence spectroscopy (Horiba), UV–Vis spectroscopy (Shimadzu), NMR (Spin-Solve).

- **Food & Biochemical Analysis**

  Antioxidant assays (Tecan), food formulations, food physicochemical analysis, complex phenolic extraction and fractionation (free, esterified, glycosylated, insoluble).

- **Microbiology & Processing**

  Microbial culture techniques, fermentation processes, sensory analysis.

- **Data Analysis & Programming**

  Statistical analysis (XLSTAT, Infostat), scientific programming (Python), basic MATLAB.

- **Professional & Transferable Skills**

  Teamwork and collaboration, leadership, problem-solving, risk analysis, conflict resolution, and scientific communication.